

Artificial Intelligence in Law

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Abstract—Artificial intelligence technologies are increasingly penetrating into various spheres of human life. Modern artificial intelligence algorithms are not capable of fully imitating legal thinking. However, artificial intelligence algorithms are capable of automating some of the routine work of lawyers. This allows professionals to deal with really difficult cases without being distracted by routine complaints. The paper discusses the use of artificial intelligence in the right.

Keywords— *Artificial Intelligence; law; information law; natural language processing*

I. INTRODUCTION

Artificial intelligence technologies are increasingly finding their application in such an area as law. All artificial intelligence algorithms that are currently used are far from real artificial intelligence. Therefore, when we talk about artificial intelligence in the context of law, it is necessary to understand that these algorithms are used only for certain functions. At the moment, a full-fledged artificial intelligence has not yet been created. Algorithms have now been developed that perform separate functions. These functions are implemented by programmers and legal experts. Modern artificial intelligence algorithms are not able to fully imitate legal thinking, since the algorithms lack logical thinking, understanding, contextual perception, and much more. Artificial intelligence techniques are machine learning techniques. At the moment, there are a number of studies that apply specific methods of machine learning. For example, the application of algorithms for natural language processing (NLP) and algorithms for deep learning. In this paper, we will look at a number of areas where machine learning techniques are applied.

II. DATA ANALYSIS

In the modern world, the volume of information is increasing. This also applies to such an area as law. Modern lawyers from various fields have to analyze large amounts of information. For example, documents of a court case, in which there are many dossiers of participants, can be analyzed using artificial intelligence algorithms and identify connections and patterns. This information can be useful to support or refute the arguments that were expressed during the trial.

The Lex Machina platform [1] allows you to predict the outcome of the case, the behavior of judges and lawyers. This platform allows you to set up unique filters and tags for a specific practice, which will allow you to find such cases and get valuable information. This saves a huge amount of time. A

very interesting function is declared on the platform's website - legal analytics. This function implies that the platform can predict winning moves for the defense or prosecution based on the study of the documents of the main case and previous similar processes. More and more experts are beginning to apply similar technologies in their practice. For example, Dr Christian Mammen (Partner, Hogan Lovells) believes that the use of tools such as Lex Machina is the key to competitiveness [2]. This suggests that many experts are beginning to think in the direction of artificial intelligence algorithms. Artificial intelligence algorithms allow you not to miss a lot of important little things in a large data stream.

Kira software [3] allows you to check contracts. This software also has the ability to learn. The system learns from the experience gained each time the final version of the contract is accepted. Users have the ability to analyze documents in 20 different languages. The system allows you to detect hidden risks in documents. It also has functionality for uploading a report on contracts. This allows the user to see all the subtle points and fix them. This system shows well how important the analysis of a large amount of data is. The more the system learns, the better it can analyze.

The project [4] allows for the analysis and due diligence of lease agreements. This software allows you to extract the necessary information from more than 50 documents in one analysis. An undeniable advantage is that the system can be trained for a specific company. Each company has its own typical document structure. Therefore, training for specific companies is about increasing the accuracy of the information extracted. This software saves you a huge amount of time and reduces the risk of missing important information.

III. FORECASTING THE RESULTS OF THE CASE

Forecasting is one of the main functions of artificial intelligence algorithms. It is already being applied in jurisprudence. This is made possible by the fact that AI algorithms can quickly process large amounts of information.

In 2014, an algorithm was created to predict the outcome of cases pending in the US Supreme Court [5]. The algorithm was able to predict the outcome of cases with 70% accuracy and 71% was the decision of individual judges. The algorithm was trained for 7 700 court decisions and over 68 000 votes of judges. Being able to predict the outcome of a case before it goes to trial is a unique opportunity. It can save time and money for both lawyers and clients. Another plus is that young

professionals with no experience can be able to analyze large flows of information and learn faster without making mistakes. Another platform Permonition [6] is a litigation database. The platform allows predicting the success of lawyers. The algorithm is based on statistics on the duration of cases and other important factors. Another useful feature is the prediction of the time for consideration of the case. This will allow the lawyer to plan his time, as well as decide whether to take on the case.

The article [7] describes an approach for predicting court decisions of the European Court of Human Rights. The approach under consideration is based on natural language processing. On average, the prediction accuracy was 79%. This is a very good result. This approach is based on solving a binary classification problem. The input data is textual content that is extracted from the case. The way out is to judge whether there has been a violation of the article of the convention on human rights. This article also contains a link to download the dataset. The dataset is in the public domain. Also, the authors of the article talk about the subsequent development of their research. They plan to focus on the thematic content of the case, as this will allow for better analysis and increase the accuracy of the algorithm's judgments.

As part of the experiment [8], the project "robot-judges" was launched in Estonia. This algorithm will have to deal with disputes in small claims of less than € 7,000. This project was launched to help judges and court clerks get rid of backlog of cases. At the first stage of the project, an algorithm for resolving contract disputes will be implemented. The algorithm will be very simple. The two parties upload documents and fill out related information. Next, the artificial intelligence algorithm makes a decision. However, this decision can be appealed to a human judge.

In 2014, a chat bot was launched [9]. This chatbot is based on artificial intelligence algorithms. For two years, according to statistics, in 2016 it was announced that the chatbot contested about 16,000 parking tickets in London and New York. At the moment, the geography of the chatbot has expanded a lot. The chatbot works with cities such as California, Buffalo (New York), Las Vegas, the University of Kansas area, Atlanta (Georgia), Salt Lake City (Utah), Rochester (New York) and many more cities. The program is very easy to use. It is installed on a mobile phone. The interface resembles a chat. The program asks a series of questions, such as whether there are clearly visible parking signs. The user is then directed to the appeal process. The appeal process fines - template. This is why artificial intelligence algorithms do the job so well. The algorithm quickly analyzes the situation and gives appropriate advice. This application is free.

IV. COMPOSING OF LEGAL DOCUMENTS

The project [10], founded by the German Institute for Artificial Intelligence, allows the drafting of lease agreements. Levertion is a patented, artificial intelligence-based data mining and contract analysis platform. What are the advantages of using this platform. Artificial intelligence algorithms extract key data from documents and then a two-stage validation of this data occurs. This saves a lot of time. This happens due to

the fact that artificial intelligence algorithms extract all the data on their own and quickly. This allows you to get rid of the routine work of managers. For example, rewriting passport numbers, place of residence and other data. It is enough just to upload documents in electronic format. The platform provides various ready-made documents, for example, a lease agreement. These agreements were drawn up by professional lawyers and there will be no pitfalls in their application. You just need to upload documents and select the necessary conditions by ticking.

There are various options for using artificial intelligence algorithms for legal purposes. For example, there is the Donna software [11]. This software is used to assist in drafting legal documents. It has a lot of interesting features. For example, checking the spelling of banknotes online. The algorithm tracks the text written by the user and immediately marks errors in the spelling of banknotes. The placeholder tool allows the user not to forget to remove comments from the document or to create an auto-substitution of the client's name so as not to be mistaken. The algorithm also tracks inconsistent formatting, broken links, vague terms, ambiguity in expressions, and more. The algorithm also tracks inconsistent formatting, broken links, vague terms, ambiguity in expressions, and more.

LegalMation [12] is a set of artificial intelligence algorithms that can automate litigation and solve routine litigation problems. The developers claim that their product is not a template generator. Artificial intelligence algorithms create legal documents at the level of a lawyer or professional lawyer. LegalMation allows you to save several days in the preparation of routine documents and save lawyers from mistakes in their preparation. The developers also state that the algorithms can be customized to generate specific documents from specific templates for a specific user. Another advantage of this product is its ease of installation and use. This reduces the time for implementation and training of employees in the company, which is an important factor.

Robot Lawyer Lisa [13] allows you to draw up a bilateral nondisclosure agreement. The principle of operation is quite simple. Two parties install this application. The initiating party answers a number of questions. The application then creates an agreement and sends it to the other party. The second party reads, makes comments. And so it continues until both sides come to an agreement. This application is very convenient. It allows you to remotely settle issues of non-disclosure of information. And it allows you to save on the use of legal aid. It is much cheaper to check the prepared document than to pay for its preparation from the very beginning.

BillyBot [14] is an application that allows you to quickly find the right attorney or mediator to solve a specific problem. It is also interesting that the algorithms first try to find a solution on free online resources. If a solution has not been found, then the selection of specialists takes place. This application saves the user money and time, since it is not always necessary to contact lawyers to find the right solution.

Automio [15] is an interactive platform for selling legal services. What is its uniqueness.

- It is possible to create a company website without programming skills.
- The ability to create surveys, questionnaires, online interviews and quizzes for clients. This makes it possible to attract customers by making them interested in an interesting presentation of the material.
- Legal Services Firm Automation. Creation of chat bots for composing simple documents.

V. AUTOMATION OF CONSIDERATION OF CASES WITH THE HELP OF ALGORITHMS OF ARTIFICIAL INTELLIGENCE

Case automation is another area where artificial intelligence algorithms can make a significant impact. Automation will allow you to process cases much faster. On the one hand, this will save customers time and money. On the other hand, this will free up time for lawyers so that they can focus on more complex cases that are still beyond the power of artificial intelligence algorithms. Another plus can be called the fact that the time of the courts is freed. Taken together, all this has very positive aspects. Automation will improve the litigation process, save inexperienced lawyers from mistakes and not miss important points when considering a case.

An interesting example is the Wevorce platform [16]. This platform allows you to find the best solution for divorce. Wevorce allows you to make the right decisions during a divorce in several stages. This minimizes spousal contact so that everything goes as smoothly as possible. If disputes arise in the decision-making process, then the system itself selects experts for consultation in a particular area of the dispute. The system offers 5 stages of the divorce process.

- Account creation. At this stage, you need to describe your situation.
- Inviting a spouse to the system.
- Remote control of all issues with expert advice.
- Obtaining a parenting plan and family agreement. These documents will be verified free of charge by a certified judge who works with this platform.
- Meet with a certified private judge and start the court session. Completion of the case is guaranteed within 30 days.

This is a great opportunity to settle all disputes peacefully and with the advice of many specialists, and also allows you to preserve the emotional state of spouses and children. It may seem that the system will eliminate practicing lawyers. This is not true. Artificial intelligence algorithms are just algorithms. They need to be verified by specialists in all fields. However, this saves a huge amount of time. When all the documents have been collected and there are recommendations of the intelligent system, the specialist can quickly grasp the essence of the issue and decide on the correctness of the algorithm. All analytics falls on artificial intelligence algorithms. This allows the expert to focus on the main points.

Another area can be called - the automation of contract agreements. PerfectNDA software [17] is a prime example.

This software allows you to speed up the process of entering into non-disclosure agreements. The software allows you to revise your own developed agreements, as well as create new ones if the company does not have one. All agreements are checked for legality. In order for the system to form an agreement, it is necessary to answer a number of questions that the system provides. The system then creates a template. The system also has a bonus - the signing of agreements using electronic signatures.

VI. INTELLECTUAL PROPERTY AUTOMATION

The process of protecting trademarks, copyrights and patents is often tedious and time-consuming work. Lawyers manually search for the necessary information. As a rule, obtaining a patent is delayed for a long time. This entails financial losses for the company. Actively evolving artificial intelligence algorithms can help accelerate this process, saving time and money.

ANAQUA Studio [18] legally uses artificial intelligence algorithms to simplify the patent drafting and filing process. The developers claim that their software can save lawyers about 20 hours when working with preliminary patent applications. And also find errors in the documentation.

SmartShell [19] software for checking, editing, formatting, and troubleshooting patent applications. The hike in this software is based on natural language processing algorithms. the user uploads the patent examiner's response to the software. Next, the text is analyzed and the answer for the expert is compiled in a custom template.

The paper [20] presents a system based on modern methods of machine learning, or rather on predictive approaches, for a detailed description of the main essence of patents. This will allow the user to better understand what a patent application consists of, what are its key features, etc.

The paper [21] discusses a gold standard approach for classifying patents and / or patent families. The approach provides design criteria and desirable attributes for assessment mechanisms. The algorithm then offers an example of a "gold standard". This article describes and discusses the results of several applications of assessment.

VII. CONCLUSION

Artificial intelligence technologies are already having a significant impact on the legal world. As technology develops, this impact will only increase. The use of artificial intelligence in legislation and legal systems helps streamline processes and save time. This allows skilled professionals to tackle more complex cases. The paper examined several examples of successful implementation of artificial intelligence algorithms in various legal fields.

ACKNOWLEDGMENT

The reported study was funded by RFBR according to the research project № 18-29-16015.

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